

Siyanda Matika

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Data and AI Lead in Johannesburg, with 8 years of experience across banking, capital markets, insurance and research. I lead data and AI initiatives from scope to production, across the full range of data: large structured datasets in SQL, Spark and Hadoop, ETL pipelines, statistical analysis, and unstructured text, image and video data, through to machine learning and LLM systems in production. In research, I collaborated on data pipelines for petabyte-scale radio astronomy data on the path to the Square Kilometre Array (SKA), as part of international science collaborations.

EXPERIENCE

Manager: AI & Data Science · Discovery Insure

Aug 2024 – present

- Own AI delivery across conversational AI, driver-coaching personalisation and the ML platform behind them. Manage the data science team and run technical hiring for the function.
- Lead the customer virtual agent for WhatsApp and web end to end: scope, agent architecture, retrieval over product documentation, guardrail strategy and integration into customer channels.
- Own the agent-routing classifier, fine-tuned on Azure OpenAI Service to read each user's intent and route the conversation to the right agent.
- Accountable for taking AI systems through privacy and security sign-off: authored the Data Protection Impact Assessments (DPIAs), coordinated multiple penetration tests with the security team, and set up data governance and adversarial testing for chatbots and LLMs.
- Established monitoring for model services on Prometheus and Grafana, with alerts on cost, warnings and errors that automatically trigger pipelines.
- Designed the Kubeflow pod infrastructure for agent development: agents run in isolated Kubernetes pods cut off from the wider network, so an agent that executes tools or code can only reach the services it is explicitly allowed to use.
- Defined the driver-coaching personalisation roadmap, moving from one fixed reward structure for every driver to coaching that adapts to each driver's behaviour.
- Co-led a transformer model on sequential health data with Discovery Health; presented it at the Discovery Actuarial and Data Science Conference, where it placed second.

Machine Learning Engineer · Navigare Securities

May 2022 – Aug 2024

- Joined as the first ML hire and worked at lead level: chose the stack, set the roadmap, and managed the AI product portfolio for capital-markets clients.
- Sat on the hiring panel for the ML team: interviewed and hired three machine learning engineers, then managed and mentored them.
- Owned GPU infrastructure and model deployment: served and fine-tuned open-weight LLMs (Llama, Mistral, Qwen), tuned serving for concurrent users, and introduced RunPod scheduling with run/sleep cycles to keep GPU spend bounded.
- Designed and ran the AWS ML stack, provisioned with Terraform: serverless model services on ECS and Fargate, CI/CD pipelines on GitHub Actions, and data pipelines on Spark and EMR with Delta Lake.
- Led conversational AI for capital-markets clients, including the decision to design an in-house hybrid retrieval framework when off-the-shelf retrieval fell short on precision.
- Conceived and led Elections Navigator, a public voter-education chatbot for South Africa's 2024 national elections, from scope to launch against the fixed election deadline.

Data Scientist · Absa Group

Jan 2020 – May 2022

- Owned the customer lifetime value model and the merchant-device cross-sell recommender from development to production; both were adopted by the marketing and cross-sell teams.
- Delivered customer spend forecasting and customer-health monitoring on Spark and Hadoop over the bank's customer data, and deployed the monitoring dashboards in Docker containers.
- Managed engagement with marketing and cross-sell stakeholders, turning model results into campaign and value-proposition recommendations presented to senior stakeholders.
- Interviewed and hired data scientists for the team, then mentored them and managed their day-to-day work.

Business Analyst · Absa Group

Jan 2017 – Jan 2018

- Owned analyses of retail banking data on trends and customer segments.
- Managed engagement with product, risk and marketing stakeholders, turning the analysis into recommendations for their teams.

SKILLS

Data engineering	SQL, Spark, Hadoop, Databricks, Delta Lake; ETL pipelines over large datasets
Statistics and analytics	Bayesian inference, time series, segmentation, forecasting, recommendation, A/B testing
Unstructured data	text, images and video: extraction, OCR, embeddings and retrieval
Machine learning	model design, training, evaluation and fine-tuning across classical ML, deep learning and transformers; PyTorch, scikit-learn, XGBoost
LLMs	open-weight models (Llama, Mistral, Qwen) and GPT models via Azure OpenAI; retrieval (RAG, hybrid search), fine-tuning (LoRA, Azure OpenAI Service), intent classification and agent routing, reinforcement learning (GRPO), guardrails
Scripting	Python, Bash, SQL
Cloud services	AWS (SageMaker, ECS, Fargate, EMR), Azure (Azure ML, App Service), GCP
GPU infrastructure and deployment	Linux GPU servers (RunPod), open-weight model serving with vLLM, concurrency tuning, Docker
MLOps and LLMOps	deployment and data pipelines, CI/CD (GitHub Actions, Azure DevOps), infrastructure as code with Terraform, Kubeflow on Kubernetes, network-isolated pods for agent workloads
Monitoring	Prometheus, Grafana, model and drift monitoring, alerts that trigger pipelines on cost, warnings and errors
Identity and access	IAM on AWS and Azure, Amazon Cognito, Azure App Service Authentication (Easy Auth)
Security and governance	Data Protection Impact Assessments (DPIAs), penetration-test reviews with security teams, adversarial testing of chatbots and LLMs, data governance
Services	FastAPI, Flask

EDUCATION

PhD research in Astrophysics, not completed · Rhodes University

2018 – 2019

Research on the Epoch of Reionization with the Hydrogen Epoch of Reionization Array (HERA), a precursor to the Square Kilometre Array (SKA).

- Collaborated on pipelines to process and analyse petabyte-scale radio interferometer data, as part of the international HERA collaboration.
- Applied Bayesian inference and Gaussian process regression to separate the faint cosmological signal from foregrounds.

Co-author of:

- Foreground modelling via Gaussian process regression: an application to HERA data, *Monthly Notices of the Royal Astronomical Society* 495(3), 2813–2826, 2020
- Detection of cosmic structures using the bispectrum phase. II. First results from application to cosmic reionization using the Hydrogen Epoch of Reionization Array, *Physical Review D* 102, 022002, 2020

MSc Astrophysics and Space Science · University of Cape Town and University of the Western Cape

2015 – 2017

Joint UCT–UWC degree through the National Astrophysics and Space Science Programme (NASSP). Thesis: *Detecting Galaxies in the Epoch of Reionization*, supervised by Mario Santos: a Bayesian framework for fitting the 21-cm absorption feature in radio galaxy spectra, to measure the redshift of galaxies in the Epoch of Reionization. Also ran SKA1-LOW noise simulations and large-scale early-universe simulations in C++.

BSc Honours Astrophysics and Space Science · University of Cape Town

2014

BSc Physics · University of the Western Cape

2011 – 2013